

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <title>Lab 8.2 - Victoria Unemployment Choropleth</title>
6   <script src="https://d3js.org/d3.v7.min.js"></script>
7   <style>
8     body {
9       font-family: 'Segoe UI', sans-serif;
10      margin: 0;
11      background-color: #f9f9f9;
12    }
13
14    h2 {
15      padding: 20px;
16      margin: 0;
17      font-size: 24px;
18    }
19
20    svg {
21      display: block;
22      margin: auto;
23      background-color: transparent;
24    }
25
26    .region {
27      stroke: #fff;
28      stroke-width: 0.5px;
29    }
30
31    .highlight {
32      stroke: black;
33      stroke-width: 2px;
34    }
35
36    .tooltip {
37      position: absolute;
38      text-align: left;
39      padding: 8px 12px;
40      font-size: 14px;
41      font-weight: 500;
42      background: #fff;
43      border: 1px solid #ccc;
44      border-radius: 4px;
45      pointer-events: none;
46      box-shadow: 0 2px 5px rgba(0,0,0,0.15);
47    }
48  </style>
49 </head>
50 <body>
51   <h2>Lab 8.2 - Victoria Unemployment Choropleth</h2>
52
53   <div id="chart"></div>
54
55   <script>
56     const width = 960;
57     const height = 600;
58
59     const svg = d3.select("#chart")
60       .append("svg")
61       .attr("width", width)
62       .attr("height", height);
63
64     const projection = d3.geoMercator()
65       .center([145, -37])
66       .translate([width / 2, height / 2])
67       .scale(5000);
68
69     const path = d3.geoPath().projection(projection);
70
71     const color = d3.scaleQuantize()
72       .range(d3.schemeBlues[7]);
73
74     const tooltip = d3.select("body").append("div")
75       .attr("class", "tooltip")
76       .style("opacity", 0);
77
78     Promise.all([
79       d3.json("IGA_VIC.json"),
80       d3.csv("VIC_IGA_unemployment.csv"),
81       d3.csv("VIC_city.csv")
82     ]).then(([geoData, unemploymentData, cityData]) => {
83       const unemploymentMap = {};
84       unemploymentData.forEach(d => {
85         unemploymentMap[d.IGA.trim().toLowerCase()] += d.unemployed;
86       });
87
88       // Set domain for color scale
89       const values = Object.values(unemploymentMap);
90       color.domain([d3.min(values), d3.max(values)]);
91
92       // Draw map
93       svg.selectAll("path")
94         .data(geoData.features)
95         .enter()
96         .append("path")
97         .attr("d", path)
98         .attr("class", "region")
99         .attr("fill", d => {
100           const name = d.properties.IGA_name.trim().toLowerCase();
101           const val = unemploymentMap[name];
102           return val ? color(val) : "#eee";
103         })
104         .on("mouseover", function (event, d) {
105           d3.select(this).classed("highlight", true);
106           const name = d.properties.IGA_name;
107           const val = unemploymentMap[name.trim().toLowerCase()];
108           tooltip.transition().duration(200).style("opacity", 1);
109           tooltip.html(`<strong>${name}</strong><br>${val ? val + ' unemployed' : 'No data'}`)
110             .style("left", (event.pageX + 10) + "px")
111             .style("top", (event.pageY - 30) + "px");
112         })
113         .on("mouseout", function () {
114           d3.select(this).classed("highlight", false);
115           tooltip.transition().duration(200).style("opacity", 0);
116         });
117
118       // Draw red circles for cities
119       svg.selectAll("circle")
120         .data(cityData)
121         .enter()
122         .append("circle")
123         .attr("cx", d => projection([d.lon, d.lat])[0])
124         .attr("cy", d => projection([d.lon, d.lat])[1])
125         .attr("r", 4)
126         .attr("fill", "red")
127         .attr("stroke", "fff")
128         .attr("stroke-width", 0.5)
129         .on("mouseover", function (event, d) {
130           d3.select(this)
131             .transition().duration(100)
132             .attr("r", 6)
133             .attr("fill", "orange");
134
135           tooltip.transition().duration(200).style("opacity", 1);
136           tooltip.html(`<strong>${d.place}</strong>`)
137             .style("left", (event.pageX + 10) + "px")
138             .style("top", (event.pageY - 30) + "px");
139         })
140         .on("mouseout", function () {
141           d3.select(this)
142             .transition().duration(100)
143             .attr("r", 4)
144             .attr("fill", "red");
145
146           tooltip.transition().duration(200).style("opacity", 0);
147         });
148     });
149   </script>
150 </body>
151 </html>

```